UNIT-I

Computer systems:

A Computer is an electronic device which performs operations such as accepts data as an input, store the data, manipulate or process the data and produce the results as an output. Main task performed by a computer

- Accept the data
- Process or manipulate the data
- Display or store the result in the form of human understanding
- Store the data, instructions and results.

A computer system consists of hardware and software.

Computer hardware is the collection of physical elements that comprise a computer system.

Computer software is a collection of computer programs and related data that provides the instructions for a computer what to do and how to do it. Software refers to one or more computer programs and data held in the storage of the computer for some purpose

Basically computer software is of three main types

System Software: System software is responsible for managing a variety of independent hardware components, so that they can work together. Its purpose is to unburden the application software programmer from the often complex details of the particular computer being used, including such accessories as communications devices, printers, device readers, displays and keyboards, and also to partition the computer's resources such as memory and processor time in a safe and stable manner.

- Device drivers
- Operating systems
- Servers
- Utilities
- Window systems

Programming Software: Programming Software usually provides tools to assist a programmer in writing computer programs, and software using different programming languages in a more convenient way. The tools include:

- Compilers
- Debuggers

- Interpreters
- Linkers
- Text editors

Application Software: Application software is developed to aid in any task that benefits from computation. It is a broad category, and encompasses Software of many kinds, including the internet browser being used to display this page. This category includes:

- Business software
- Computer aided design
- Databases
- Decision making software
- Educational software
- Image editing

Computer Generations

Let us now discuss the development in Computer Technology over the different generations.

1) First Generation

The period 1940 to 1956, roughly considered as the First Generation of Computer. The first generation computers were developed by using vacuum tube or thermionic valve machine.The input of this system was based on punched cards and paper tape; however, the output was displayed on printouts. The first generation computers worked on binary-coded concept (i.e., language of 0-1). **Examples:** ENIAC, EDVAC, etc.

2) Second Generation

The period 1956 to 1963 is roughly considered as the period of Second Generation of Computers.The second generation computers were developed by using transistor technology. In comparison to the first generation, the size of second generation was smaller. In comparison to computers of the first generation, the computing time taken by the computers of the second generation was lesser

3) Third Generation

The period 1963 to 1971 is roughly considered as the period of Third Generation of computers.The third generation computers were developed by using the Integrated Circuit (IC) technology.In comparison to the computers of the second generation, the size of the computers of the third generation was smaller.In comparison to the computers of the second generation, the computing time taken by the computers of the third generation was lesser.The third generation computer

consumed less power and also generated less heat.The maintenance cost of the computers in the third generation was also low.The computer system of the computers of the third generation was easier for commercial use.

4) Fourth Generation

The period 1972 to 2010 is roughly considered as the fourth generation of computers.The fourth generation computers were developed by using microprocessor technology.By coming to fourth generation, computer became very small in size, it became portable.The machine of fourth generation started generating very low amount of heat.It is much faster and accuracy became more reliable.The production cost reduced to very low in comparison to the previous generation.It became available for the common people as well.

5) Fifth Generation

The period 2010 to till date and beyond, roughly considered as the period of fifth generation of computers.By the time, the computer generation was being categorized on the basis of hardware only, but the fifth generation technology also included software.The computers of the fifth generation had high capability and large memory capacity.Working with computers of this generation was fast and multiple tasks could be performed simultaneously.Some of the popular advanced technologies of the fifth generation include Artificial intelligence, Quantum computation, Nanotechnology, Parallel processing, et

Computer Generations

Generation	Hardware Components	Characteristics	Computers
First Generation (1942-1959)	 Vaccum Tubes	Machine Language Huge Size Highly Expensive High Consumption of Electricity	ENIAC UNIVAC EDVAC EDSAC IBM-701
Second Generation (1959-1965)	 Transistors Magnetic Tapes	Batch processing, Multiprogramming OS Expensive FORTRAN, COBOL	IBM 7000 CDC 1604 ATLAS NCR 304 Honeywell 400
Third Generation (1965-1975)	 Integrated Circuits	Remote processing, time-sharing, Multiprogramming OS Faster, Compact & Cheaper PASCAL PL/1, BASIC, ALGOL-68	IBM 360/370 PDP 8/11 CDC 6600
Fourth Generation (1975-1988)	 VLSI Microprocessor circuits	Time-sharing, real-time networks, distributed, GUI OS Faster, Compact & Affordable C, C++, DBASE	DEC 10 STAR 1000 CRAY-1/11 Apple II VAX 9000
Fifth Generation (1988-Present)	 ULSI Microprocessor circuits	Parallel Processing & Artificial Intelligence technology C and C++, Java, .Net	IBM Pentium Param

Storage Devices

The storage unit is a part of the computer system which is employed to store the information and instructions to be processed. A storage device is an integral part of the computer hardware which stores information/data to process the result of any computational work. Without a storage device, a computer would not be able to run or even boot up. Or in other words, we can say that a storage device is hardware that is used for storing, porting, or extracting data files. It can also store information/data both temporarily and permanently.

Types of Computer Memory

1. Primary Memory
2. Secondary Memory
3. Tertiary Memory

1. Primary Memory: It is also known as internal memory and main memory. This is a section of the CPU that holds program instructions, input data, and intermediate results. It is generally smaller in

size. RAM (Random Access Memory) and ROM (Read Only Memory) are examples of primary storage.

2. Secondary Memory: Secondary storage is a memory that is stored external to the computer. It is mainly used for the permanent and long-term storage of programs and data. Hard Disks, CDs, DVDs, Pen/Flash drives, SSD, etc, are examples of secondary storage.

3. Tertiary Memory: Tertiary Memory is a type of Memory that is rarely used in personal computers and due to this, tertiary memory is not considered to be an important one. Tertiary memory works automatically without human intervention.

Types of Computer Storage Devices

Now we will discuss different types of storage devices available in the market. These storage devices have their own specification and use. Some of the commonly used storage devices are:

1. Primary Storage Devices
2. Magnetic Storage Devices
3. Flash memory Devices
4. Optical Storage Devices
5. Cloud and Virtual Storage

1. **Primary Storage Devices**

- **RAM**: It stands for Random Access Memory. It is used to store information that is used immediately or we can say that it is a temporary memory. Computers bring the software installed on a hard disk to RAM to process it and to be used by the user. Once, the computer is turned off, the data is deleted. With the help of RAM, computers can perform multiple tasks like loading applications, browsing the web, editing a spreadsheet, experiencing the newest game, etc. It allows you to modify quickly among these tasks, remembering where you're in one task once you switch to a different task. It is also used to load and run applications, like your spreadsheet program, answers commands, like all edits you made within the spreadsheet, or toggle between multiple programs, like once you left the spreadsheet to see the email. Memory is nearly always actively employed by your computer. It ranges from 1GB – 32GB/64GB depending upon the specifications. There are different types of RAM, and although they all serve the same purpose, the most common ones are :

1)SRAM: It stands for Static Random Access Memory. It consists of circuits that retain stored information as long as the power supply is on. It is also known as volatile memory. It is used to build Cache memory. The access time of SRAM is lower and it is much faster as compared to DRAM but in terms of cost, it is costly as compared to DRAM.

2) **DRAM:** It stands for Dynamic Random Access Memory. It is used to store binary bits in the form of electrical charges that are applied to capacitors. The access time of DRAM is slower as compared to SRAM but it is cheaper than SRAM and has a high packaging density.

3) SDRAM: It stands for Synchronous Dynamic Random Access Memory. It is faster than DRAM. It is widely used in computers and others. After SDRAM was introduced, the upgraded version of double data rate RAM, i.e., DDR1, DDR2, DDR3, and DDR4 was entered into the market and widely used in home/office desktops and laptops.

- **ROM:** It stands for Read-Only Memory. The data written or stored in these devices are non-volatile, i.e, once the data is stored in the memory cannot be modified or deleted. The memory from which will only read but cannot write it. This type of memory is non-volatile. The information is stored permanently during manufacture only once. ROM stores instructions that are used to start a computer. This operation is referred to as bootstrap. It is also used in other electronic items like washers and microwaves. ROM chips can only store a few megabytes (MB) of data, which ranges between 4 and 8 MB per ROM chip. There are two types of ROM:

1) **PROM:** PROM is Programmable Read-Only Memory. These are ROMs that can be programmed. A special PROM programmer is employed to enter the program on the PROM. Once the chip has been programmed, information on the PROM can't be altered. PROM is non-volatile, that is data is not lost when power is switched off.
2) **EPROM:** Another sort of memory is the Erasable Programmable Read-Only Memory. It is possible to erase the info which has been previously stored on an EPROM and write new data onto the chip.

3) **EEPROM:** EEPROM is Electrically erasable programmable read-only memory. Here, data can be erased without using ultraviolet light, with the use of just applying the electric field.

2. Magnetic Storage Devices

- **Floppy Disk:** Floppy Disk is also known as a floppy diskette. It is generally used on a personal computer to store data externally. A Floppy disk is made up of a plastic cartridge and secured with a protective case. Nowadays floppy disk is replaced by new and effective storage devices like USB, etc.
- **Hard Disk:** Hard Disk is a storage device (HDD) that stores and retrieves data using magnetic storage. It is a non-volatile storage device that can be modified or deleted n number of times without any problem. Most computers and laptops have HDDs as their secondary storage device. It is actually a set of stacked disks, just like phonograph records. In every hard disk, the data is recorded electromagnetically in concentric circles or we can say track present on the hard disk, and with the help of a head just like a phonograph arm(but fixed in a position) to read the information present on the track. The read-write speed of HDDs is not so fast but decent. It ranges from a few GBs to a few and more TB.
- **Magnetic Card:** It is a card in which data is stored by modifying or rearranging the magnetism of tiny iron-based magnetic particles present on the band of the card. It is also known as a swipe card. It is used like a passcode(to enter the house or hotel room), credit card, identity card, etc.
- **Tape Cassette:** It is also known as a music cassette. It is a rectangular flat container in which the data is stored in an analog magnetic tape. It is generally used to store audio recordings.
- **SuperDisk:** It is also called LS-240 and LS-120. It is introduced by Imation Corporation and it is popular with OEM computers. It can store data up to 240 MB.

3. Flash Memory Devices

It is a cheaper and more portable storage device. It is the most commonly used device to store data because is more reliable and efficient as compared to other storage devices. Some of the commonly used flash memory devices are:

- **Pen Drive:** It is also known as a USB flash drive that includes flash memory with an integrated USB interface. We can directly connect these devices to our computers and laptops and read/write data into them in a much faster and more efficient way. These devices are very portable. It ranges from 1GB to 256GB generally.
- **SSD:** It stands for Solid State Drive, a mass storage device like HDD. It is more durable because it does not contain optical disks inside like hard disks. It needs less power as compared to hard disks, is lightweight, and has 10x faster read and writes speed as compared to hard disks. But, these are costly as well. While SSDs serve an equivalent function as hard drives, their internal components are much different. Unlike hard drives, SSDs don't have any moving parts and thus they're called solid-state drives. Instead of storing data on magnetic platters, SSDs store data using non-volatile storage. Since SSDs haven't any moving parts, they do not need to "spin up". It ranges from 150GB to a few more TB.
- **SD Card:** It is known as a Secure Digital Card. It is generally used with electronic devices like phones, digital cameras, etc. to store larger data. It is portable and the size of the SD card is also small so that it can easily fit into electronic devices. It is available in different sizes like 2GB, 4GB, 8GB, etc.
- **Memory Card:** It is generally used in digital cameras. printers, game consoles, etc. It is also used to store large amounts of data and is available in different sizes. To run a memory card on a computer you require a separate memory card reader.
- **Multimedia Card:** It is also known as MMC. It is an integrated circuit that is generally used in-car radios, digital cameras, etc. It is an external device to store data/information.

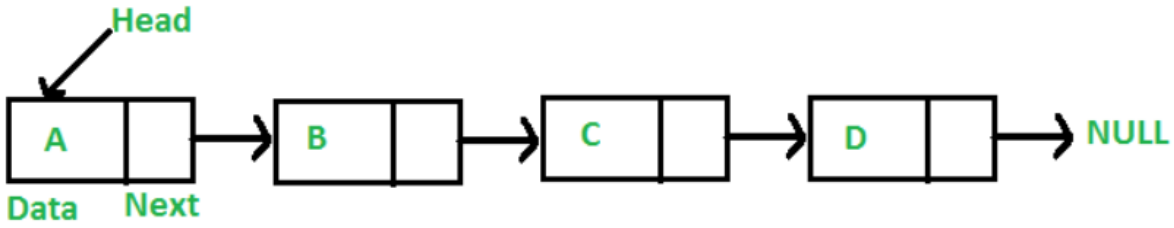
4. Optical Storage Devices

Optical Storage Devices is also secondary storage device. It is a removable storage device. Following are some optical storage devices:

- **CD:** It is known as Compact Disc. It contains tracks and sectors on its surface to store data. It is made up of polycarbonate plastic and is circular in shape. CD can store data up to 700MB. It is of two types:
    - **CD-R:** It stands for Compact Disc read-only. In this type of CD, once the data is written can not be erased. It is read-only.
    - **CD-RW:** It stands for Compact Disc Read Write. In this type of CD, you can easily write or erase data multiple times.
- **DVD:** It is known as Digital Versatile Disc. DVDs are circular flat optical discs used to store data. It comes in two different sizes one is 4.7GB single-layer discs and another one is 8.5GB double-layer discs. DVDs look like CDs but the storage capacity of DVDs is more than as compared to CDs. It is of two types:
    - **DVD-R:** It stands for Digital Versatile Disc read-only. In this type of DVD, once the data is written can not be erased. It is read-only. It is generally used to write movies, etc.
    - **DVD-RW:** It stands for Digital Versatile Disc Read Write. In this type of DVD, you can easily write or erase data multiple times.
- **Blu-ray Disc:** It is just like CD and DVD but the storage capacity of blu ray is up to 25GB. To run a Blu-ray disc you need a separate Blu-ray reader. This Blu-ray technology is used to read a disc from a blue-violet laser due to which the information is stored in greater density with a longer wavelength.

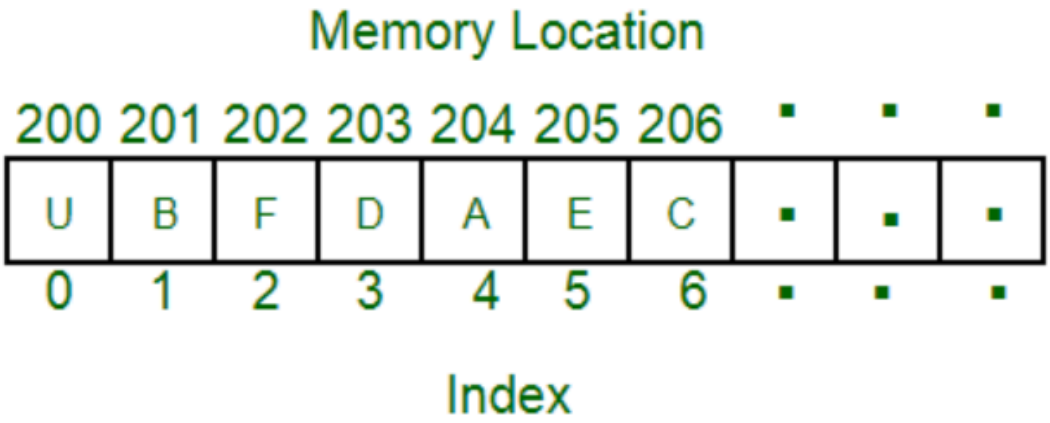
These are 4 types of memory access methods:

1.Sequential Access:- In this method, the memory is accessed in a specific linear sequential manner, like accessing in a single Linked List. The access time depends on the location of the data.



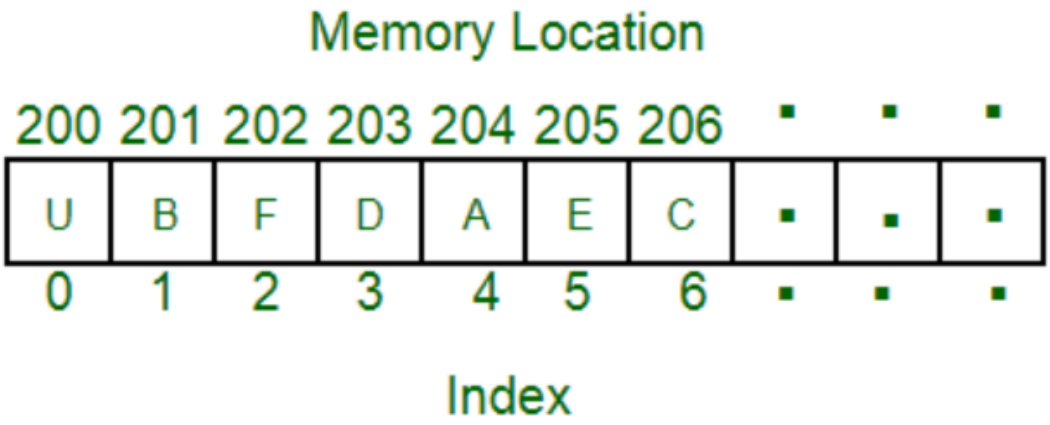
Applications of this sequential memory access are magnetic tapes, magnetic disk and optical memories.

2. Random Access: In this method, any location of the memory can be accessed randomly like accessing in Array. Physical locations are independent in this access method.



Applications of this random memory access are RAM and ROM.

3. Direct Access: In this method, individual blocks or records have a unique address based on physical location. access is accomplished by direct access to reach a general vicinity plus sequential searching, counting or waiting to reach the final destination. This method is a combination of above two access methods. The access time depends on both the memory organization and characteristics of storage technology. The access is semi-random or direct.



Application of thus direct memory access is magnetic hard disk, read/write header.

4. Associate Access: In this memory, a word is accessed rather than its address. This access method is a special type of random access method. Application of thus Associate memory access is Cache memory.

COMPUTER LANGUAGES

In order to communicate with the computer user also needs to have a language that should be understood by the computer. For this purpose, different languages are developed for performing different types of work on the computer. Basically, languages are divided into two categories according to their interpretation.

1. Low Level Languages.

2. High Level Languages.

Low Level Languages

Low level computer languages are machine codes or close to it. Computer cannot understand instructions given in high level languages or in English. It can only understand and execute instructions given in the form of machine language i.e. language of 0 and 1. There are two types of low level languages:

- Machine Language.
- **Assembly Language**

Machine Language: It is the lowest and most elementary level of Programming language and was the first type of programming language to be Developed. Machine Language is basically the only language which a computer Can understand. In fact, a manufacturer designs a computer to obey just one Language, its machine code, which is represented inside the computer by a String of binary digits (bits) 0 and 1. The symbol 0 stands for the absence of Electric pulse and 1 for the presence of an electric pulse . Since a computer is Capable of recognizing electric signals, therefore, it understand machine Language.

Advantages of Machine Language

i) It makes fast and efficient use of the computer.

ii) It requires no translator to translate the code i.e. Directly understood by the computer

Disadvantages of Machine Language:

i) All operation codes have to be remembered

ii) These languages are machine dependent i.e. a particular Machine language can be used on only one type of computer **Assembly Language**

It was developed to overcome some of the many inconveniences of machine language. This is another low level but a very important language in which operation codes and operands are given in the form of alphanumeric symbols instead of 0‟s and l‟s. These alphanumeric symbols will be known as mnemonic codes and can have maximum up to 5 letter combination e.g. ADD for addition, SUB for subtraction, START,LABEL etc. Because of this feature it is also known as „Symbolic Programming Language‟. This language is also very difficult and needs a lot of practice to master it because very small English support is given to this language. The language mainly helps in compiler orientations. The instructions of the Assembly language will also be converted to machine codes by language translator to be executed by the computer.

Advantages of Assembly Language

i) It is easier to understand and use as compared to machine language.

ii) It is easy to locate and correct errors.

iii) It is modified easily

Disadvantages of Assembly Language

i) Like machine language it is also machine dependent.

ii) Since it is machine dependent therefore programmer Should have the knowledge of the hardware also.

High Level Languages

High level computer languages give formats close to English language and the purpose of developing high level languages is to enable people to write programs easily and in their own native language environment (English). High-level languages are basically symbolic languages that use English words and/or mathematical symbols rather than mnemonic codes. Each instruction in the high level language is translated into many machine language instructions thus showing one-to-many translation

Types of High Level Languages

Many languages have been developed for achieving different variety of tasks, some are fairly specialized others are quite general purpose.

These are categorized according to their use as

a) Algebraic Formula-Type Processing. These languages are oriented towards the computational procedures for solving mathematical and statistical problem

Examples are

- BASIC (Beginners All Purpose Symbolic Instruction Code).
- FORTRAN (Formula Translation).
- PL/I (Programming Language, Version 1).
- ALGOL (Algorithmic Language).
- APL (A Programming Language).

b) Business Data Processing:

These languages emphasize their capabilities for maintaining data processing procedures and files handling problems. Examples are:

- COBOL (Common Business Oriented Language).
- RPG (Report Program Generator

c) String and List Processing: These are used for string manipulation including search for patterns, inserting and deleting characters. Examples are:

- LISP (List Processing).
- Prolog (Program in Logic).

d) Object Oriented Programming Language

In OOP, the computer program is divided into objects. Examples are:

· C++

· **Java**

e) Visual programming language: these are designed for building Windows-based applications Examples are:

· Visual Basic

· Visual Java

· Visual C

Advantages of High Level Language

Following are the advantages of a high level language:

- · User-friendly
- · Similar to English with vocabulary of words and symbols
- · Therefore it is easier to learn.
- · They are easier to maintain.

Disadvantages of High Level Language

- A high-level language has to be translated into the machine language by a translator and thus a price in computer time is paid.
- The object code generated by a translator might be inefficient Compared to an equivalent assembly language program

Creating and Running Programs:

There are four steps in this process.

1. Writing and editing the program using Text editor (source code).
2. Compile the program using any C compiler.(.bak file)
3. Linking the program with the required library modules(object file)
4. Executing the program. (.Exe file)

Creating and Editing a C Program in C Programming Language compiler: Writing or creating and editing source program is a first step in c language. Source code is written in c programming language according to the type of problem or requirement, in any text editor.

Saving C Program in C Programming Language: Source code is saved on the secondary storage. Source code is saved as text file. The extension of file must be ".c". Example the file name is "learn c programming language.c"

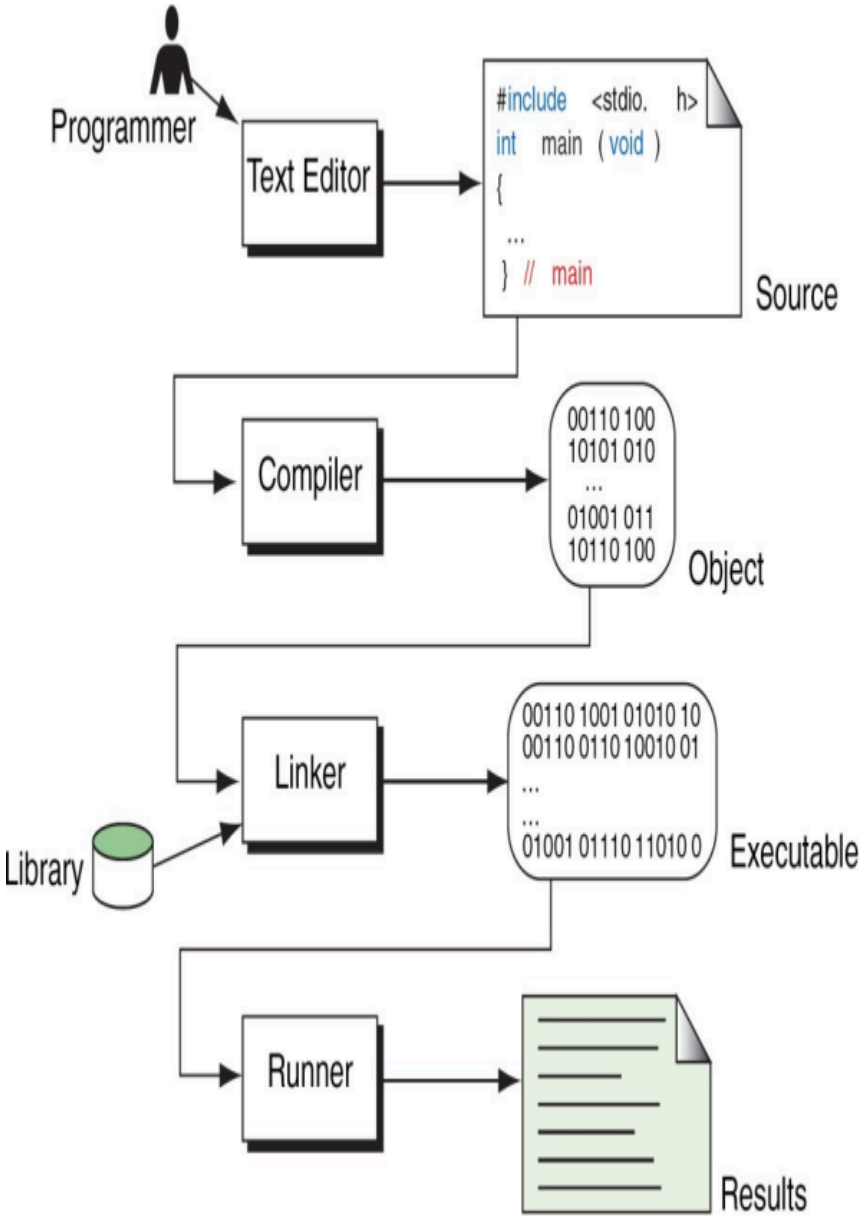
Compiling C program in C Programming Language: Computer does not understand c programming language. It understands only 0 and 1 means machine language. So c programming language code is converted into machine language. The process of converting source code in to machine code is called compiling.

Compiler is a program that compiles source code. Compiler also detects errors in the source program. If compiling is successful, the source program is converted into an object program. Object program is saved on disk. The extension of file is ".obj"

Linking in C programming Language: There are many built in functions available in c programming language. These functions are also called library functions. These functions are stored in different header files.

Loading program: The process of transferring a program from secondary storage to main memory for execution is called loading a program. A program called loader does loading.

Executing program: Execution is the last step. In this step program starts execution. Its instructions start working and output of the program display on the screen.



Types of program errors

We distinguish between the following types of errors:

1. **Syntax errors**: errors due to the fact that the syntax of the language is not respected.
2. **Semantic errors**: errors due to an improper use of program statements.
3. **Logical errors**: errors due to the fact that the specification is not respected.

Syntax errors occur when a programming language's rules are not followed, while semantic errors occur when statements are not meaningful:

Syntax errors: These errors occur when a compiler or interpreter is unable to understand a source code statement. Syntax errors cannot be undetected because the program will not run if they exist.

Semantic errors: These errors occur when program statements are used incorrectly. Semantic errors can go undetected for years because the results may seem reasonable. However, they can be harmful or even deadly if they involve financial transactions or medical equipment.

Here are some examples of syntax and semantic errors:

- Syntax error: int a b is a syntax error because it is missing a semicolon
- Semantic error: X*Y = Z is a semantic error because the statements are not meaningful
- Semantic error in C#: int x = 12.3; is a semantic error because 12.3 is not an integer literal

Types of Translators

Here are some types of translators in computer language and examples of them:

1. Interpreter

The translation of a single statement of the source program into machine code is done by a language processor and executes immediately before moving on to the next line is called an interpreter. If there is an error in the statement, the interpreter terminates its translating process at that statement and displays an error message. The interpreter moves on to the next line for execution only after the removal of the error. An Interpreter directly executes instructions written in a programming or scripting language without previously converting them to an object code or machine code. An interpreter translates one line at a time and then executes it.

Example: Perl, Python and Matlab.

2. Assembler: The Assembler is used to translate the program written in Assembly language into machine code. The source program is an input of an assembler that contains assembly language instructions. The output generated by the assembler is the object code or machine code understandable by the computer. Assembler is basically the 1st interface that is able to communicate humans with the machine. We need an assembler to fill the gap between human and machine so that they can communicate with each other. code written in assembly language is some sort of mnemonics(instructions) like ADD, MUL, MUX, SUB, DIV, MOV and so on. and the assembler is basically able to convert these mnemonics in binary code. Here, these mnemonics also depend upon the architecture of the machine.

For example, the architecture of intel 8085 and intel 8086 are different.

3.**Compiler:** The language processor that reads the complete source program written in high-level language as a whole in one go and translates it into an equivalent program in machine language is called a Compiler. Example: C, C++, C#.In a compiler, the source code is translated to object code successfully if it is free of errors. The compiler specifies the errors at the end of the compilation with line numbers when there are any errors in the source code. The errors must be removed before the compiler can successfully recompile the source code again the object program can be executed number of times without translating it again.

Algorithms

An **algorithm** is a step-by-step procedure or formula for solving a problem. It takes an input, processes it through well-defined steps, and produces an output. Algorithms are fundamental to computer science and software engineering, and they are used in everything from basic programs to complex systems like artificial intelligence.

Characteristics of a Good Algorithm

1. **Input**: An algorithm should have clearly defined inputs.
2. **Output**: It should produce an output after execution.
3. **Definiteness**: Each step of the algorithm should be clear and unambiguous.
4. **Correctness**: The algorithm should correctly solve the problem.
5. **Finiteness**: The algorithm must terminate after a finite number of steps.
6. **Efficiency**: It should consume a reasonable amount of resources (time and space).
7. **Generality**: It should be applicable to a wide range of problems.

Example:

Algorithm for Adding Two Numbers

1. **Start**.
2. **Input**: Read two numbers, A and B.
3. **Process**: Calculate the sum S = A + B.
4. **Output**: Display the result S.
5. **End**.

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Flowchart In Programming

A flowchart is a diagrammatic representation of an algorithm. A flowchart can be helpful for both writing programs and explaining the program to others.

Symbols Used In Flowchart

Symbol	Purpose	Description
	Flow line	Indicates the flow of logic by connecting symbols.
	Terminal(Stop/Start)	Represents the start and the end of a flowchart.
	Input/Output	Used for input and output operation.
	Processing	Used for arithmetic operations and data-manipulations.
	Decision	Used for decision making between two or more alternatives.

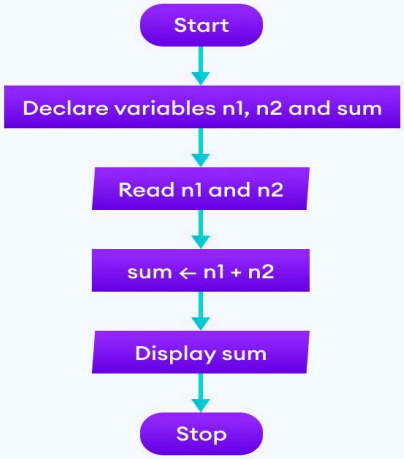


On-page Connector

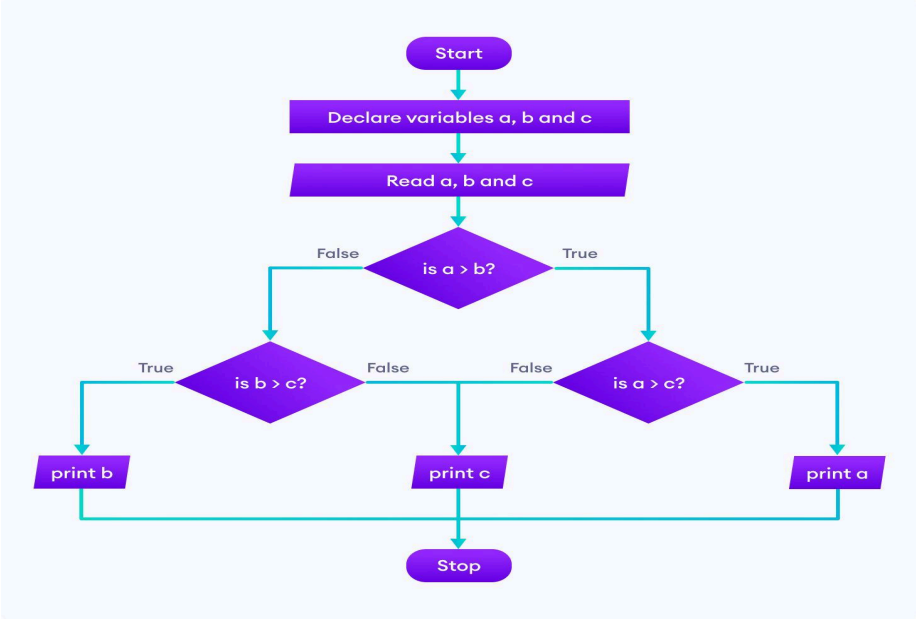
Used to join different flowline

Examples of flowcharts in programming

1. Add two numbers entered by the user.



2. Find the largest among three different numbers entered by the user.



Pseudocode

The pseudocode in C is an informal way of writing a program for better human understanding. It is written in simple English, making the complex program easier to understand. Pseudocode cannot be compiled or interpreted.

Example:

```
if "1"
   print response
      "I am case 1"
if "2"
   print response
      "I am case 2"
```

Difference between Algorithm and Pseudocode

Algorithm	Pseudocode
An Algorithm is used to provide a solution to a particular problem in form of a well-defined step-based form.	A Pseudocode is a step-by-step description of an algorithm in code-like structure using plain English text.

Algorithm	Pseudocode
An algorithm only uses simple English words	Pseudocode also uses reserved keywords like if-else, for, while, etc.
These are a sequence of steps of a solution to a problem	These are fake codes as the word pseudo means fake, using code like structure and plain English text
There are no rules to writing algorithms	There are certain rules for writing pseudocode
Algorithms can be considered pseudocode	Pseudocode cannot be considered an algorithm
It is difficult to understand and interpret	It is easy to understand and interpret

Difference between Flowchart and Pseudocode

Flowchart	Pseudocode
A Flowchart is pictorial representation of flow of an algorithm.	A Pseudocode is a step-by-step description of an algorithm in code like structure using plain English text.
A Flowchart uses standard symbols for input, output decisions and start stop statements. Only uses different shapes like box, circle and arrow.	Pseudocode uses reserved keywords like if-else, for, while, etc.
This is a way of visually representing data, these are nothing but the graphical representation of the algorithm for a better understanding of the code	These are fake codes as the word pseudo means fake, using code like structure but plain English text instead of programming language
Flowcharts are good for documentation	Pseudocode is better suited for the purpose of understanding